There are 8 pages containing examples of results and visualizations that I hope to generate.

RESULTS

3. Client-Side Results

3a. Local Feature Selection or Ranking. Present top-N features for each client with visualizations such as:

- Feature importance bar plots for each client based on local model.
- Heatmaps of selected features across clients.
- Correlation plots.

(Optional, any consistency across different feature selection methods (if multiple)?)

Exemplary visualization:

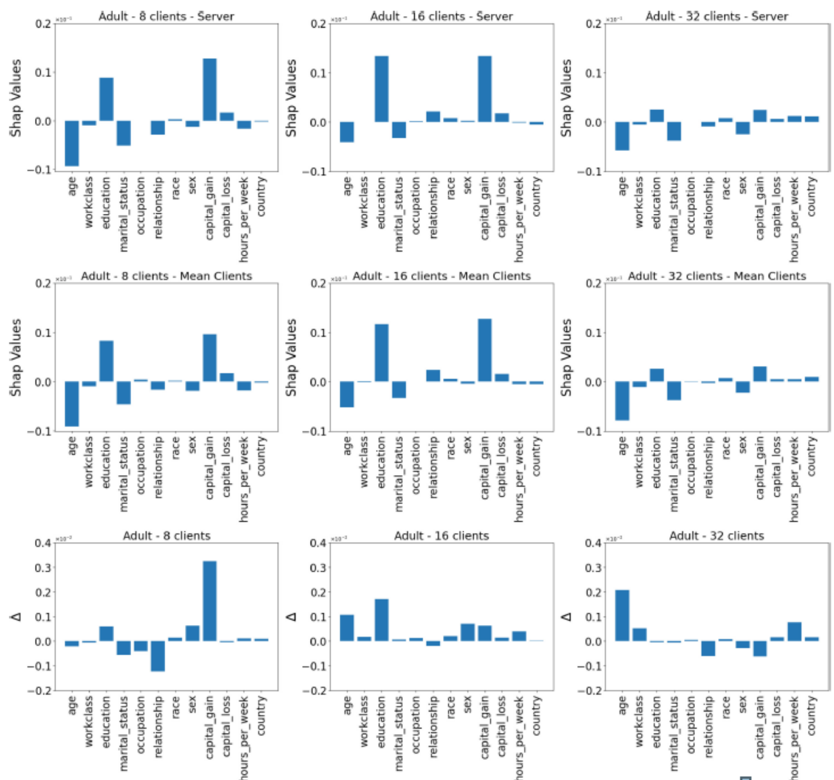

Fig. 4. SHAP values for Adult. Top: calculated by the server. Middle: calculated by the clients. Bottom: Difference between SHAP values obtained by the server and those obtained by the clients.

From: Luca Corbucci, "Explaining Black-Boxes in Federated Learning"
https://www.dropbox.com/scl/fi/c70g773hgk0quf4azohkp/1-SHAp-for-client-n-server-feature-explanatibility.pdf?rlkey=hk60gl9o5qtijdsk8mdodholx&dl=0

3b. Local Model Performance & Visualizations

Report for each client / each model:

- Table for the train/test performance (mean ± SD across folds or repeated runs) and suggested plots:
  - ROC curves / PR curves.
  - Regression scatter plots (predicted vs. actual).
  - Confusion matrices.

  Exemplary visualization:

Table 6: Performance of Individual Client Models

atase t	Accuracy				Optimizatio n Method
	Epoch -20	Epoch -30	Epoch -40	Epoch -50	
Client 1	0.9333	0.9478	0.9572	0.9672	Random Forest with Bayesian Optimization
Client 2	0.9549	0.9623	0.9683	0.9735	SVM with Random Search
Client 3	0.9740	0.9792	0.9824	0.9862	Gradient Boosting with PSO

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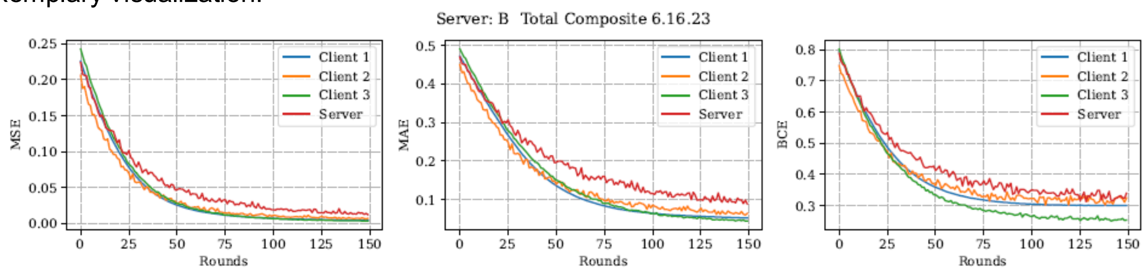
  -
  - https://www.dropbox.com/scl/fi/bp9v044fv04cggf35sh70/000-Exact-flow-and-topics-an-results.pdf?rlkey=d98bcm87gedmx7gh7vz2nbdm8&dl=0

4 Federated (Server-Side) Results

4a Federated Model Performance. Present results with**:**

- Table of train/test performance (mean ± SD).
- Performance change across communication rounds (learning curve).
- Comparison charts: federated vs. local model performance.

Exemplary visualization:



- Figure: Comparison of federated vs local model training performance. (Add validation performance as well).

Exemplary visualization:

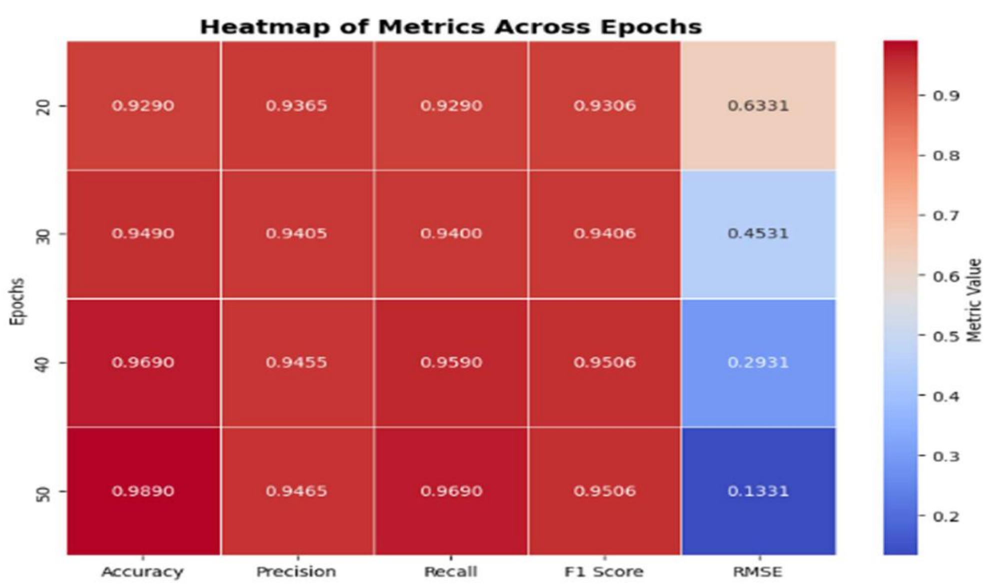


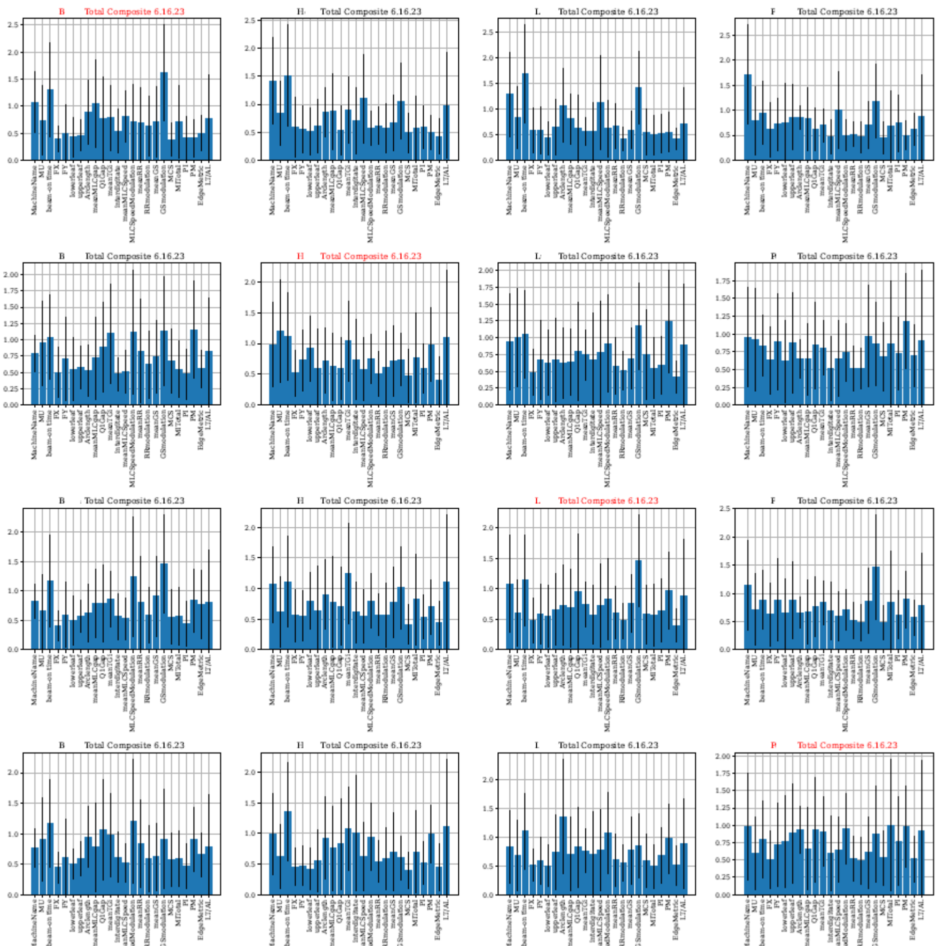
Figure 9: Heatmap of Metrics Across Epochs of Aggregation Model

- https://www.dropbox.com/scl/fi/bp9v044fv04cggf35sh70/000-Exact-flow-and-topics-an-results.pdf?rlkey=d98bcm87gedmx7gh7vz2nbdm8&dl=0

4b Federated Feature Selection or Ranking. Present results with **(**suggested plots):

- Global feature importance (rank list) bar chart.
- Performance-over-rounds curve.
- Overlap analysis (Venn diagrams) between federated and local features.

Exemplary visualization:



4c Payoff (contribution) from each client towards the Federated model :

Exemplary visualization:

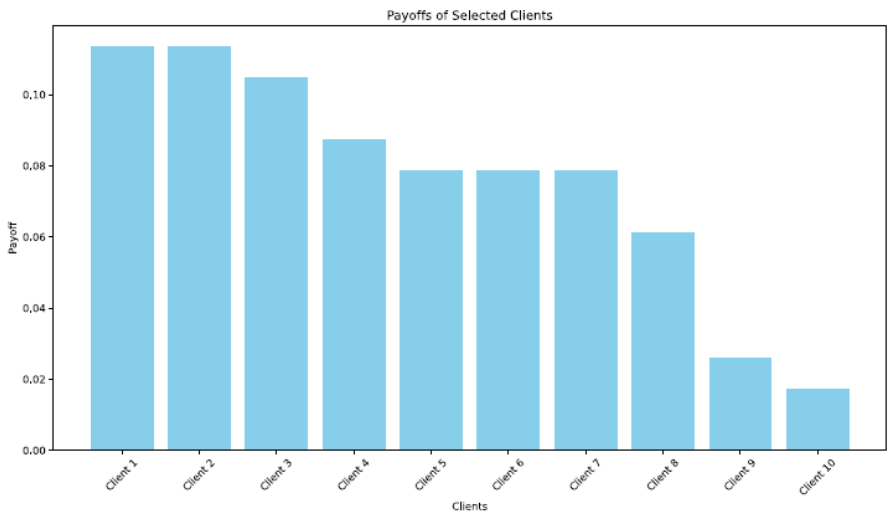


FIGURE 9. Payoff values of each client.

Y. SUPRIYA AND RAJESWARI CHENGODEN, "Breast Cancer Prediction Using Shapely and Game Theory in Federated Learning Environment" https://www.dropbox.com/scl/fi/gyaxj75ruj4qmrswgckum/1-Shapely_and_acc-vs-rounds-plots.pdf?rlkey=f7vesn4bhpqr4sq1obp9hwn0x&dl=0

4d Payoff /fitness score from each client towards the Federated model :

Exemplary visualization:

Datasets	Naming	Selected Features	Fitness Scores
Client-1	C1-F1	Relation	-0.206240
	C1-F2	raisedhands	-0.093552
	C1-F3	VisITedResources	0.005105
	C1-F4	AnnouncementsView	0.050640
	C1-F5	Discussion	-0.055448
	C1-F6	StudentAbsenceDays	-0.072508
Client-2	C2-F1	Age(Years)	0.056823
	C2-F2	Time spent on social media (Hours)	0.008703
	C2-F3	Average marks scored before pandemic in traditional classroom	-0.111603
	C2-F4	Your interaction in online mod	0.070353
	C2-F5	Performance in online	0.115048
Client-3	C3-F1	StudentID	-0.126501
	C3-F2	GPA	-0.960180

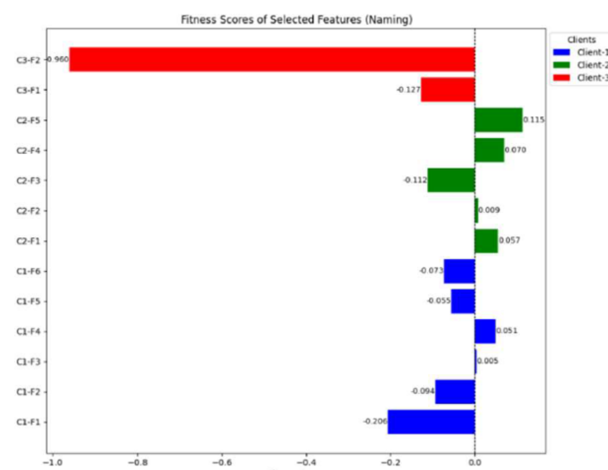


Figure 5: Visualization of Fitness Scores for Selected Features Across Clients with Unique Naming

https://www.dropbox.com/scl/fi/bp9v044fv04cggf35sh70/000-Exact-flow-and-topics-an-results.pdf?rlkey=d98bcm87gedmx7qh7vz2nbdm8&dl=0

4e Comparison of local vs federated training parameters:

Table 1. Training parameter configurations

Parameter	Sect. 4.1	Sect. 4.2	Sect. 4.3
FL parameters			
Training iterations	1000	–	2000
Clients per iteration	3	48	48–432
Weight update method	Mean, weighted by the local dataset size		
Local training parameters			
Epochs	100	10	
Batch size	10	64	
Training algorithm	SGD	Adadelta	
Loss function	Sparse categorical crossentropy		
Learning rate	0.005	–	

https://www.dropbox.com/scl/fi/s5h02r6dj1csjksbf9qjf/1-Training-parameter-summary.pdf?rlkey=8qjwvlxaewucw33l41iip2rku&dl=0

5. Validation: Applying Federated Model Back to Each Client

After the federated model & feature set are optimized:

5a Cross-Client Validation Using Global Features

For each client:

- Apply federated-selected features (using each client's local feature values).
- Evaluate performance using the global federated model.
- Compare:
  - **Performance using federated model + global features**
  - **Performance using local model + local features** (obtained from step 3b above)

  Suggested plots- Bar chart comparing:
  - Local model accuracy vs. federated model (client-specific predictions).
  - Scatter plots of regression predictions.
- Summary performance table.
- Statistical significance testing (optional, e.g., paired t-test to test if difference b/w federated and local model are significant).

Exemplary visualization:

Table 10. Statistical significance of TreeNetX compared to baseline models

Client	Metric	Comparison model	Mean (Tree NeTX ± SD)	Mean (baseline ± SD)	Test used	P-value	Effect size (Cohen's D)	95% CI for mean difference	Stat. significant (A = 0.05)
Client-1	Accuracy	CatBoost	0.892 ± 0.015	0.874 ± 0.017	McNemar	0.041	0.45 (medium)	[0.003, 0.035]	Yes
Client-1	F1 Score	CatBoost	0.887 ± 0.018	0.872 ± 0.019	Paired t-test	0.048	0.40 (small–medium)	[0.001, 0.029]	Yes
Client-1	AUC	CatBoost	0.941 ± 0.020	0.933 ± 0.021	Paired t-test	0.067	0.25 (small)	[–0.002, 0.019]	No
Client-2	Accuracy	LightGBM	0.905 ± 0.010	0.892 ± 0.012	McNemar	0.028	0.55 (medium)	[0.004, 0.021]	Yes
Client-2	F1 Score	LightGBM	0.912 ± 0.013	0.881 ± 0.015	Paired t-test	0.012	0.70 (medium–large)	[0.010, 0.045]	Yes
Client-2	AUC	LightGBM	0.963 ± 0.009	0.942 ± 0.011	Paired t-test	0.009	0.80 (large)	[0.012, 0.038]	Yes
Client-3	Accuracy	XGBoost	0.883 ± 0.014	0.876 ± 0.016	McNemar	0.059	0.30 (small)	[–0.001, 0.017]	No
Client-3	F1 Score	XGBoost	0.872 ± 0.016	0.860 ± 0.018	Paired t-test	0.073	0.28 (small)	[–0.002, 0.024]	No
Client-3	AUC	XGBoost	0.957 ± 0.011	0.936 ± 0.013	Paired t-test	0.018	0.65 (medium–large)	[0.006, 0.036]	Yes

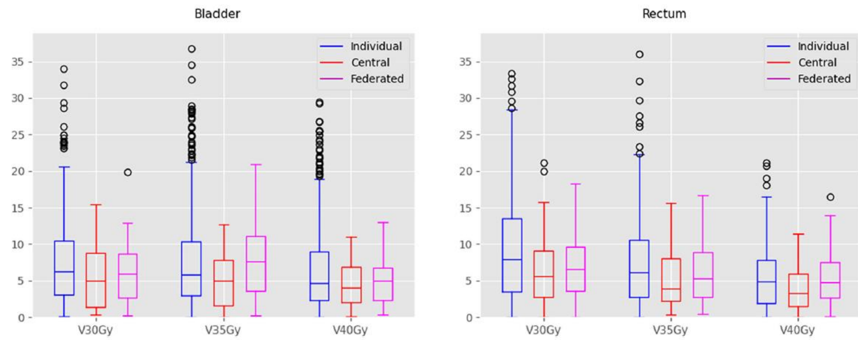
https://www.dropbox.com/scl/fi/6yhibkgegy99u1ax0lmo9/000-Exact-flow-and-topics-FEDERATED-TREE-BASED-ENSEMBLES-WITH-SHAP-for-Lung.pdf?rlkey=aw208wjycrqk2hvgcq9xjipfx&dl=0

6. Optional but Valuable Additions

Ablation Studies

- 6a. Compare individual client model vs. federated vs. centralized training (if data sharing allowed as a simulation).

  Exemplary visualization:



- **FIGURE 7** Prediction absolute errors of individual models, central models, and federated models for other key metrics, including V_{30Gy}, V_{35Gy}, and V_{40Gy} for both bladder and rectum.

  From : https://www.dropbox.com/scl/fi/8p364ln1osvb1fb39m63v/0_most-relevanat-figs-_2024-MPhys-FL-for-enhanced-DVH-parameter-predict.pdf?rlkey=gmq2yfu3wzoro7lb0m76lylzj&dl=0

- 6b. Compare performance with different numbers of global features.

Table 3: Comparison of Experimental Results Between Single and Multiple Clients

Dataset	Experiment Setup	Number of Clients	Selected Features	Accuracy	F1 Score
Iris	Single without FS	1	4	0.8889	0.8669
	Single with FS	1	2	0.9556	0.9484
	Multiple without FS	3	4	0.8667	0.8464
	Multiple with FS	3	2	0.9556	0.9484
Breast Cancer	Single without FS	1	30	0.9532	0.9497
	Single with FS	1	15	0.9298	0.9241
	Multiple without FS	5	30	0.9883	0.9873
	Multiple with FS	5	16	0.9708	0.9685
Digits	Single without FS	1	64	0.9111	0.9098
	Single with FS	1	37	0.9167	0.9160
	Multiple without FS	6	64	0.9481	0.9462
	Multiple with FS	6	32	0.9333	0.9304
Wine	Single without FS	1	13	0.9444	0.9421
	Single with FS	1	7	0.9630	0.9611
	Multiple without FS	3	13	0.9630	0.9633
	Multiple with FS	3	7	0.9815	0.9804
Palmer Penguins	Single without FS	1	6	0.9615	0.9572
	Single with FS	1	4	0.9231	0.9137
	Multiple without FS	3	6	0.9904	0.9882
	Multiple with FS	3	4	0.9712	0.9655
Seeds	Single without FS	1	7	0.9000	0.8841
	Single with FS	1	4	0.8833	0.8551
	Multiple without FS	3	7	0.9000	0.8814
	Multiple with FS	3	5	0.9333	0.9222
WineQuality	Single without FS	1	12	0.9826	0.9774
	Single with FS	1	5	0.9615	0.9497
	Multiple without FS	6	12	0.9872	0.9832
	Multiple with FS	6	5	0.9646	0.9573
Digit Recognizor	Single without FS	1	784	0.8957	0.8943
	Single with FS	1	331	0.8952	0.8933
	Multiple without FS	10	784	0.9096	0.9082
	Multiple with FS	10	325	0.9033	0.9019

Chi Zhang “A Federated Learning based Online Feature Selection Algorithm for Streaming Data”
https://www.dropbox.com/scl/fi/fj9v4yimetojh2ujcsvxf/1-good-description-of-flow-feat-select-from-local-n-global-n-online-dynam.pdf?rlkey=umcpldv3hejhsd27vou7d5cjg&dl=0